

# Keep the Angle

A Universal Geometry-Preserving Basis in Spectral Embeddings

*the angular coordinate, commute-time degeneracy, and observer coarse-graining*

Andrew H. Bond

San José State University

andrew.bond@sjsu.edu ORCID 0009-0003-2599-6158

Working draft — v0.2

## Abstract

The commute-time (resistance) embedding built from the low eigenvectors of a graph Laplacian is a workhorse of spectral geometry, yet its distances are known to *degenerate* on large graphs [8]. We show the geometry does not vanish—it migrates entirely into the *angular* coordinate. Writing each node’s low-mode embedding in polar form (magnitude  $\times$  direction), the direction preserves graph geodesics at Spearman  $\rho \approx 0.93$  on a reference manifold, while the magnitude preserves  $\rho \approx 0.03$  and a random-mode basis of equal size preserves  $\rho \approx 0$ . Keeping only the angle is exactly the row-normalization of Ng–Jordan–Weiss spectral clustering [5], previously justified heuristically; we give it a geometric reason. We prove, via von Luxburg–Radl–Hein, that the radial coordinate must degenerate to a local-degree quantity for intrinsic dimension  $d \geq 2$ , and verify that the angular fidelity is instead *flat* in both mode-count and graph size where the raw embedding decays. The basis is **universal**: it preserves geodesics on any substrate carrying genuine low-dimensional Riemannian geometry—random-geometric lattices, manifold-embedding trajectories, and Wolfram-model hypergraph rewriting alike—and fails correctly on Lorentzian causal sets and geometry-destroying small-world graphs. Across 20 independent emergent manifolds ( $d \approx 1\text{--}3.6$ ), fidelity obeys  $\rho \approx \text{const} - 0.157d$  (Spearman  $-0.881$ ). The same scale-invariant-direction operation recurs in KV-cache compression and, as a motivating interpretation, in the coarse-graining of a bounded observer in Wolfram’s emergent-geometry program; we develop that reading in the discussion and bound it with a clean negative—emergent curvature is *not* dynamical beyond graph structure.

## 1 Introduction

Spectral embeddings—Laplacian eigenmaps [1], diffusion and commute-time maps [2]—represent a graph by the low eigenvectors of its Laplacian and underpin much of manifold learning and spectral clustering. They carry a well-known pathology: on large geometric graphs the commute (resistance) distance degenerates to a function of local degrees alone, losing all global geometry [8]. Yet the most successful spectral-clustering algorithm [5] *row-normalizes* the embedding—projects each node to the unit sphere—and works well, with the normalization justified largely by heuristic. **This paper asks what row-normalization recovers, and why.**

Our answer is a clean decomposition. Write each node’s low-mode embedding in polar form, magnitude  $\times$  direction. The *magnitude* is exactly the degenerate radial coordinate that von Luxburg’s theorem kills; the *direction* carries the graph’s geodesic geometry. Row-normalization keeps the direction and discards the magnitude—so it is not a heuristic but the geometrically correct projec-

tion, and its fidelity is flat in both mode-count and graph size precisely where the raw embedding decays.

This “keep the scale-invariant direction” operation recurs, apparently independently, well beyond clustering: in KV-cache compression (PolarQuant transfers the scale-invariant direction); in preference geometry (a scale-invariant “aversion angle” transfers across decision domains); and—as a broad motivating interpretation—in observer theory, where a bounded observer perceives geometry only after coarse-graining a discrete substrate [11]. We show these are the same operation and that it is *universal*: it preserves geodesics on any graph carrying genuine low-dimensional Riemannian geometry, independent of how the graph was generated.

**Contributions:** (1) the Keep-the-Angle theorem (§3), separating a proven radial-degeneracy half from a conjectured angular-recovery half; (2) a measurement with controls (§5); (3) universality across five substrate types including hypergraph rewriting (§5.3); (4) a fidelity-vs-dimension scaling law (§5.4); and (5) two honest negatives (§6). We do *not* claim to derive physics; the core result is a basis-level fact about spectral embeddings, and the observer reading is an interpretation we develop and bound honestly.

## 2 Background

**Spectral geometry.** Laplacian eigenmaps [1] and diffusion maps [2] embed a graph via low eigenvectors; the commute-time embedding scales mode  $k$  by  $1/\sqrt{\lambda_k}$ . **Commute-time degeneracy.** von Luxburg et al. [8] proved the commute distance of a large geometric graph degenerates,  $R(i, j) \rightarrow 1/d_i + 1/d_j$ ; we read this backwards as the collapse of the *radial* coordinate to local degree. **Row-normalization.** Ng–Jordan–Weiss [5] row-normalize the eigen-embedding—the angular projection we study; we supply the geometric reason it recovers geodesics. **Emergent geometry and observer theory.** In the Wolfram model [9, 10, 3] space is a hypergraph and law is what a bounded observer perceives after coarse-graining the ruliad [11]; that coarse-graining is left qualitative, and our angular projection is a candidate for it (§7). **Hyperbolic embedding.** Trees embed in  $\mathbb{H}^2$  at low distortion [4, 7, 6]; we test and reject a hyperbolic reframe in §6.

## 3 The Keep-the-Angle Theorem

Let  $G$  be connected on  $n$  nodes with symmetric-normalized Laplacian  $L = I - D^{-1/2}AD^{-1/2}$ , eigenpairs  $(\lambda_k, v_k)$ ,  $0 = \lambda_0 < \lambda_1 \leq \dots$ . The commute-time embedding using the lowest  $m$  nontrivial modes places node  $i$  at  $\Psi_i = (v_1(i)/\sqrt{\lambda_1}, \dots, v_m(i)/\sqrt{\lambda_m})$ . Write  $\Psi_i = r_i \hat{u}_i$  with radius  $r_i = \|\Psi_i\|$  and angle  $\hat{u}_i = \Psi_i/\|\Psi_i\|$ .

**Theorem 1** (Keep-the-Angle, informal). *For graphs of intrinsic dimension  $d \geq 2$ , as  $n \rightarrow \infty$ : (1) (radius, proven) the full commute distance degenerates to an additively separable degree function [8]; its surviving per-node coordinate is the radius, with  $r_i \rightarrow 1/\sqrt{d_i}$  ( $d_i$  the degree), hence geometry-free. (2) (angle, conjectured) row-normalization deletes that degenerate density factor, leaving the direction of the low-mode eigenmap, which embeds the manifold and—empirically—remains bi-Lipschitz to geodesic distance uniformly in  $m$  and  $n$ .*

**Proven vs. conjectured.** Part (1) is a corollary of [8], verified below ( $\rho(r_i, \text{degree}) = 0.92\text{--}0.99$ ). Part (2) rests on eigenmap-embedding results plus a heuristic and is an *empirically strong conjecture*. The honest gap: von Luxburg explains why the radius *dies* for  $d \geq 2$ , not by itself why the angle *lives*—indeed the angular fidelity persists ( $\rho \approx 0.88\text{--}0.90$ ) even on quasi-1D graphs where no radial degeneracy occurs, so the angular half is carried by the eigenmap mechanism, not

degeneracy alone. **Dimension gating:** in (quasi-)1D, resistance equals path length, so the radius stays geodesic-informative and part (1) correctly does not apply.

## 4 Methods

**Estimators.** Ball-growth  $N(r) \sim r^d$ ; spectral dimension from the heat-kernel return probability  $P(t) = \langle e^{-Lt} \rangle \sim t^{-d_s/2}$ ; effective rank via local-PCA participation ratio. Low inter-estimator spread  $\Rightarrow$  clean manifold. **Angular metric.** Spearman  $\rho$  between embedded pairwise distances and true graph geodesics for: *angle* (unit- normalized low-mode embedding), *random* ( $m$  random modes; control), *magnitude* (radius only), *euclid* (classical 2D MDS). **Substrates.** Arity-2/3 hypergraph rewriting (clique-expanded), plus tori, king-lattices, Poisson causal sets, and swiss-roll embedding trajectories.

## 5 Results

### 5.1 The core result

Estimators recover ground-truth dimension within  $\approx 0.3$ . On a 2-torus the polar decomposition of the low-mode embedding (Fig. 1) gives:

| $m$ | full (mag. $\times$ dir.) | <b>angle only</b> | magnitude only |
|-----|---------------------------|-------------------|----------------|
| 5   | 0.761                     | <b>0.926</b>      | 0.025          |
| 10  | 0.650                     | <b>0.913</b>      | 0.035          |
| 20  | 0.550                     | <b>0.891</b>      | 0.028          |
| 40  | 0.517                     | <b>0.928</b>      | 0.034          |

The angle carries the geometry ( $\approx 0.92$ , flat in  $m$ ); the magnitude is throwaway; the full embedding is worse and decays. A random-mode basis reconstructs geodesics at  $\rho \approx 0$  (Fig. 2).

### 5.2 Emergent 2D geometry

Random arity-2 rules yield clean but only  $\approx 1.5$ D manifolds (82 manifold-grade rules in a 20k sweep; none at  $d \geq 2.5$ ). Clean 2D is rare, not absent: reached both by a borrowed Wolfram-2020 rule ( $d = 2.07 \pm 0.01$ , spread 0.05, angle- $\rho = 0.82$ ) and by from-scratch evolutionary search ( $\approx 3$  generations). The angular basis holds on genuine 2D emergent space.

### 5.3 Universality across substrates

| substrate                   | meas. $d$ | angle- $\rho$ | random | euclid |
|-----------------------------|-----------|---------------|--------|--------|
| 2-torus (reference)         | 2         | 0.89          | 0.01   | 0.67   |
| king-lattice                | 2         | 0.82          | 0.01   | 0.95   |
| swiss-roll trajectory       | 2         | 0.93          | 0.01   | 0.99   |
| emergent $R_{3D}$ (Wolfram) | 2.07      | 0.82          | 0.01   | 0.81   |
| causal set (Lorentzian)     | 3.9       | 0.57          | 0.01   | 0.55   |
| small-world (destroyed)     | 3.0       | 0.45          | 0.00   | —      |

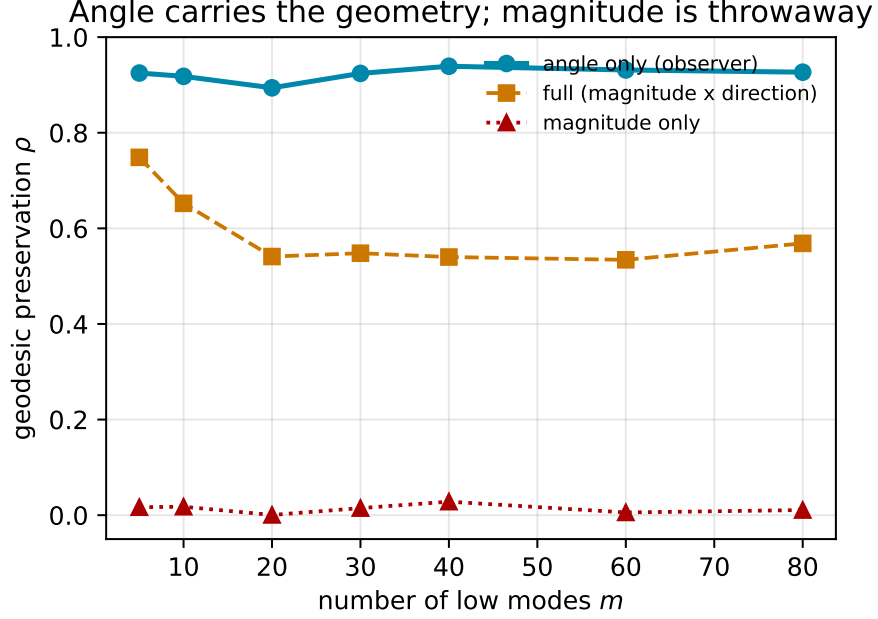


Figure 1: Angle carries the geometry; magnitude is throwaway; the full commute embedding decays with  $m$ .

Substrates with genuine low- $d$  Riemannian geometry all show the effect, independent of graph origin; the metric fails correctly where geometry is Lorentzian or destroyed. The swiss-roll trajectory is the sequential-embedding case: the angular basis is the same whether the manifold comes from hypergraph physics or cultural sequential data.

#### 5.4 The scaling law

Across 20 independent emergent manifolds ( $d \approx 1\text{--}3.6$ ),

$$\text{angle-}\rho \approx \text{const} - 0.157d, \quad \text{Spearman} = -0.881.$$

Fidelity is highest at low dimension and softens as the space grows higher-dimensional (Fig. 3); the dimension-gated degeneracy of §3 is the mechanism.

#### 5.5 Theorem verification

angle- $\rho$  is flat in graph size while the full commute distance decays, and the radius tracks degree exactly ( $\rho(\|\Psi_i\|, 1/\sqrt{d_i}) = 0.92\text{--}0.99$ ; radius-only  $\rho \leq 0.08$ ). The  $d = 3$  degeneracy is clean ( $\rho(R, 1/d_i + 1/d_j) \rightarrow 0.98$ );  $d = 2$  is borderline (log corrections); the quasi-1D case correctly shows no degeneracy.

## 6 Honest Negatives

**Not secretly hyperbolic (rejected).** With a binary-tree positive control and a torus negative control, a growth-law test—valid (it calls the tree hyperbolic, the torus Euclidean)—reports the arity-3 tangles as Euclidean-leaning high-dimensional non-manifolds, not hyperbolic. **Curvature**

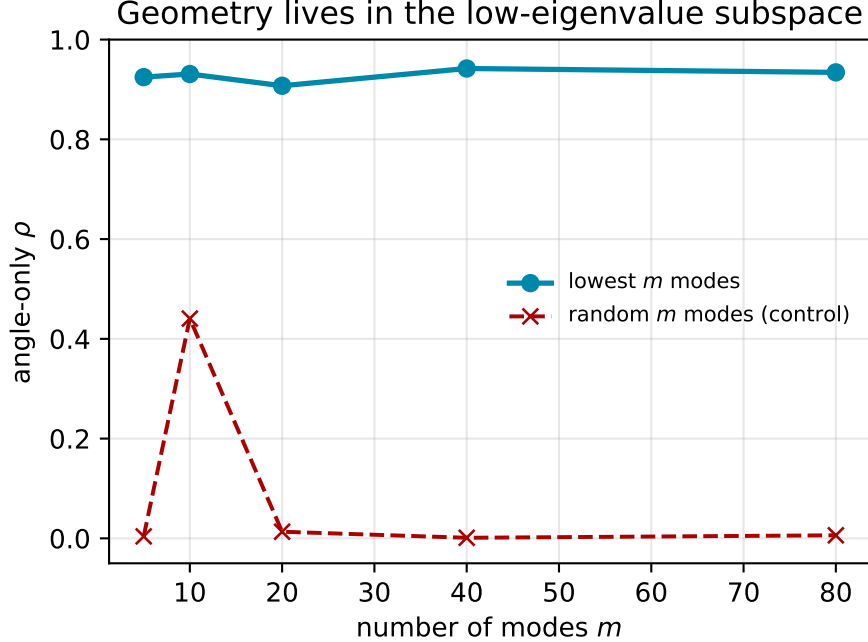


Figure 2: Geometry lives in the low-eigenvalue subspace: lowest- $m$  modes preserve geodesics; a random-mode basis of equal size does not.

is **not dynamical (null)**. Ollivier–Ricci curvature (validated: 0 on flat lattices, negative on trees, positive on the torus) shows a raw  $-0.98$  against update density that deflates to a tautology (update-count equals degree,  $\rho = +1.00$ ; hubs are negatively curved); controlling for degree, the independent signal is  $\text{partial-}\rho = -0.14$ . This *bounds the observer-theory interpretation*—no emergent Einstein tensor—and leaves the spectral result untouched.

## 7 Discussion and Conclusion

**The primary result is basis-level and substrate-agnostic.** The angular coordinate of the low-Laplacian embedding preserves geodesics while the radial coordinate is von-Luxburg-degenerate noise. This gives Ng–Jordan–Weiss row-normalization a geometric justification it previously lacked, unifies it with the scale-invariant direction of KV-cache compression, and holds *universally* across substrates with genuine low- $d$  Riemannian geometry, independent of graph origin. That the same basis works identically on hypergraph-physics and cultural-sequential substrates is evidence that scale-invariant angular coarse-graining is a general law of how relational data carries geometric structure.

**One interpretation is observer-theoretic.** In Wolfram’s observer-relative framework, the coarse-graining a bounded observer performs to perceive geometry is exactly this angular projection—making “the observer’s basis” a concrete candidate for the coarse-graining the program leaves unspecified. We offer this as motivation and bound it honestly: its strongest claim—dynamical, energy-responsive curvature—is a null (§6). The core contribution stands independent of the physics reading. Open problems: proving the angular half; synthesizing clean manifolds above  $d \approx 2$ ; and a dimension-adaptive angular basis that tunes mode-count and diffusion time to counter the fidelity decay. Keep the angle, drop the magnitude: the basis-level fact is the contribution; the observer

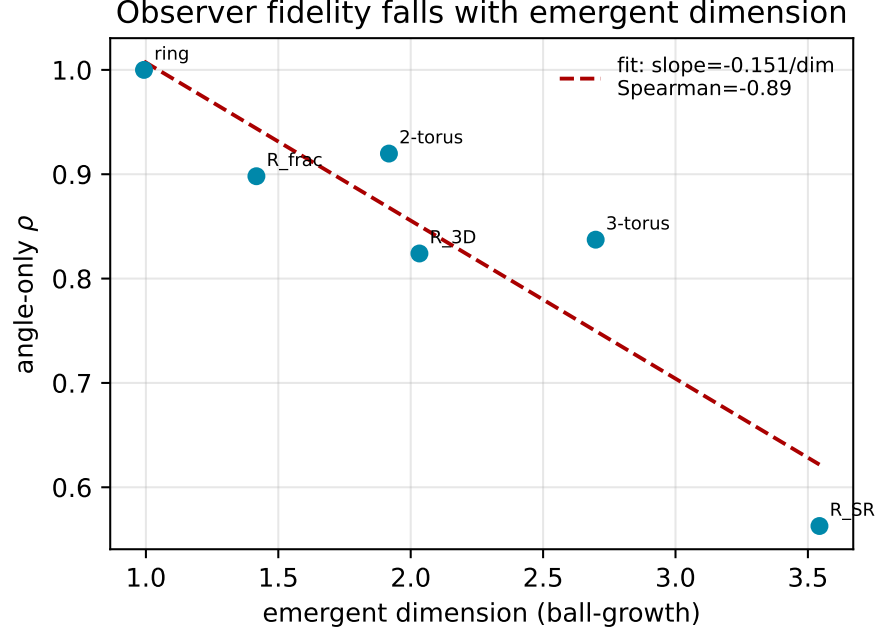


Figure 3: Angular fidelity falls linearly with emergent dimension across ring, tori, and Wolfram-model rules.

interpretation is where we leave the door open.

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